

13 Show the score

The function `display_score()` instructs the score turtle to rewrite the score, putting the latest total on the screen. This function is called whenever the caterpillar reaches a leaf.

```
def display_score(current_score):
    score_turtle.clear()
    score_turtle.penup()
    x = (t.window_width() / 2) - 50
    y = (t.window_height() / 2) - 50
    score_turtle.setpos(x, y)
    score_turtle.write(str(current_score), align='right', \
                       font=('Arial', 40, 'bold'))
```

50 pixels from the top

50 pixels from the right

14 A new leaf

When a leaf is reached, the function `place_leaf()` is called to move the leaf to a new, random location. It chooses two random numbers between -200 and 200. These numbers become the x and y coordinates for the next leaf.

```
def place_leaf():
    leaf.ht()
    leaf.setx(random.randint(-200, 200))
    leaf.sety(random.randint(-200, 200))
    leaf.st()
```

ht is short for hideturtle.

Chooses random coordinates to move the leaf.

st is short for showturtle.

15 Turning the caterpillar

Next, to connect the keyboard keys to the caterpillar, add four new direction functions after the `start_game()` function. To make this game a little trickier, the caterpillar can only make 90-degree turns. As a result, each function first checks to see which way the caterpillar is moving before altering its course. If the caterpillar's going the wrong way, the function uses `setheading()` to make it face the right direction.

```
game_over()
break

def move_up():
    if caterpillar.heading() == 0 or caterpillar.heading() == 180:
        caterpillar.setheading(90)

def move_down():
    if caterpillar.heading() == 0 or caterpillar.heading() == 180:
        caterpillar.setheading(270)

def move_left():
    if caterpillar.heading() == 90 or caterpillar.heading() == 270:
        caterpillar.setheading(180)

def move_right():
    if caterpillar.heading() == 90 or caterpillar.heading() == 270:
        caterpillar.setheading(0)
```

Check if the caterpillar is heading left or right.

A heading of 270 sends the caterpillar down the screen.